



DFDOF

Forensic Baseline Report

Drone Forensic Decision and Orchestration Framework

Case Identifier:	ft-df050-2018-android
Operator:	Floris
Analysis Started:	2026-06-18 12:37:30
Analysis Completed:	2026-06-18 12:48:11
Evidence Directory:	C:\Users\Floris\Documents\dji_drones\train_mavic_air\df050\2018\android
Output Directory:	C:\Users\Floris\Documents\dji_drones\train_mavic_air\df050\2018\dfdof_output\ft-df050-2018-android
Evidence Sources:	3
Pipeline Version:	DFDOF v1.0 (proof of concept)

This report is an automated forensic baseline produced by DFDOF. It is not a court-ready expert report. All findings are bounded by the data available and must be interpreted by a qualified forensic examiner. The populated case output directory is the authoritative source and provides the full overview of the pipeline's findings.

1. Evidence Intake and Chain of Custody

The following input evidence files were ingested. SHA-256 and SHA-1 hashes were computed at intake to establish integrity baselines. No modification to original evidence files is performed by DFDOF.

File	Classification	Acquisition	Size	Hash Timestamp
android_logical.zip	controller_android	logical	4.9 GB	2026-06-18 12:37:58
df047_external.E01	drone_sd	physical	22.7 GB	2026-06-18 12:39:12
internal_dji_export.zip	drone_flight_storage	logical	1.9 GB	2026-06-18 12:39:18

Evidence Source Path

C:\Users\Floris\Documents\dji_drones\train_mavic_air\df050\2018\android

Cryptographic Hashes

android_logical.zip		controller_android	logical
SHA-256:	dacc6f652a337e0a0c672c28500c94b28e838d2006bef66856dffe3408fd2e8		
SHA-1:	136d2c44d1883c44dba0e2267e95773e99b568cb		
df047_external.E01		drone_sd	physical
SHA-256:	e15d62a740b9b9f270867fdd6326a3f09beade45c00f51a01d69b46f5d416602		
SHA-1:	2afa47a8d984989b82629f228f96c5fe238a5aaa		
internal_dji_export.zip		drone_flight_storage	logical
SHA-256:	c3aedebb87d9f35bd1b3170970d20cc9ef5d4eea1ea64246310844de85fafba4		
SHA-1:	84e0e21ec32712fe648ec890065e4fc0e0d9ed5f		

2. Pipeline Execution Summary

Completed Phases

1. p1_provenance
2. p2_image_parsing
3. p3_artefact_extraction
4. p4_decision_and_orchestration
5. p5_normalisation_and_anomaly_checking
6. p6_multisource_correlation
7. p7_analysis_and_validation
8. p8_automated_reporting

Anomaly Flags

1. [p2 - controller android - device and backup info]: DeviceInfo.xml not found for android_logical.zip
2. [p2 - controller android - device and backup info]: ApplicationInfo.xml not found for android_logical.zip
3. [p2 - controller ios]: source evidence not found
4. [p3 - controller android - drone logs]: no artefacts found
5. [p3 - controller ios]: source evidence not found
6. [p4 - controller android - account data]: account data parse failed for dji.go.v4.xml
7. [p4 - controller ios]: source evidence not found
8. [p4 - drone flight storage - drone logs]: DatCon reported an error processing: FLY001_1.DAT
9. [p4 - drone flight storage - drone logs]: DatCon reported an error processing: FLY002_1.DAT
10. [p4 - drone flight storage - drone logs]: DatCon reported an error processing: FLY004_1.DAT
11. [p4 - drone flight storage - drone logs]: DatCon reported an error processing: FLY005_1.DAT
12. [p4 - drone flight storage - drone logs]: DatCon reported an error processing: FLY007_1.DAT
13. [p4 - drone flight storage - drone logs]: DatCon reported an error processing: FLY009_1.DAT
14. [p4 - drone flight storage - drone logs]: DatCon reported an error processing: FLY010_1.DAT
15. [p4 - drone flight storage - drone logs]: DatCon reported an error processing: FLY011_1.DAT
16. [p4 - drone flight storage - drone logs]: DatCon reported an error processing: FLY013_1.DAT
17. [p4 - drone flight storage - drone logs]: DatCon reported an error processing: FLY014_1.DAT
18. [p4 - drone flight storage - drone logs]: ExtractDJI only produced GIMBAL processing: DJI_ASSISTANT_EXPORT_FILE_2018-05-07_16-25-09.DAT
19. [p4 - drone flight storage - drone logs]: ExtractDJI only produced GIMBAL processing: DJI_ASSISTANT_EXPORT_FILE_2018-05-07_16-26-14.DAT
20. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY003_1.csv
21. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY006_1.csv
22. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY008_1.csv
23. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY012_1.csv
24. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY015_1.csv
25. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY016_1.csv
26. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY017_1.csv
27. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY018_1.csv

Tool Execution Status

Tool	Invocations	Successes	Status
datcon	18	8	[~] Partial
exiftool	20	20	[OK]
extractdji	3	1	[~] Partial
fls	1	1	[OK]
icat	16	16	[OK]
mmls	1	1	[OK]
parser_android	1	1	[OK]
txtlogtocsvtool	3	3	[OK]

Processing Coverage Score

Metric	Result
Evidence Sources Detected	3 / 4
Artefact Categories With Data	7 / 7
Flights With Primary Correlation	0 / 3
Tools Succeeded	6 / 8

3. Device and Account Identification

Field	Value	Confidence	Sources
Installed DJI App(s)	dji.go.v4	[K] Controller-only	1
Drone Serial Number	0K1DF2P1BD 0K1DF2P1BD65JJ	[I] Inconsistent	3
Drone Model	Mavic Air	[K] Controller-only	3
DJI Application Version	4.2.8	[K] Controller-only	3

4. Flight Analysis - Flight 01

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Flight Record SHA-256: bffdddec1cca18d5d1ce30da8d55225f92205d755dd6f883305a43df8bfc5fbf1
Flight Record Name: DJIFlightRecord_2018-04-19_[12-46-27].csv
Start: 2018-04-19T18:46:27.485000+00:00
End: 2018-04-19T18:47:11.822000+00:00
Duration: 44.3 s

Flight Metrics

Peak Height: 4.9 m
Distance Travelled: 7.89 m
Video Recording: Yes
Photo Capture: No
Photos Taken: 0
Recording Duration: 44.0 s
Record Complete: Yes

Fly Modes Observed (Chronological)

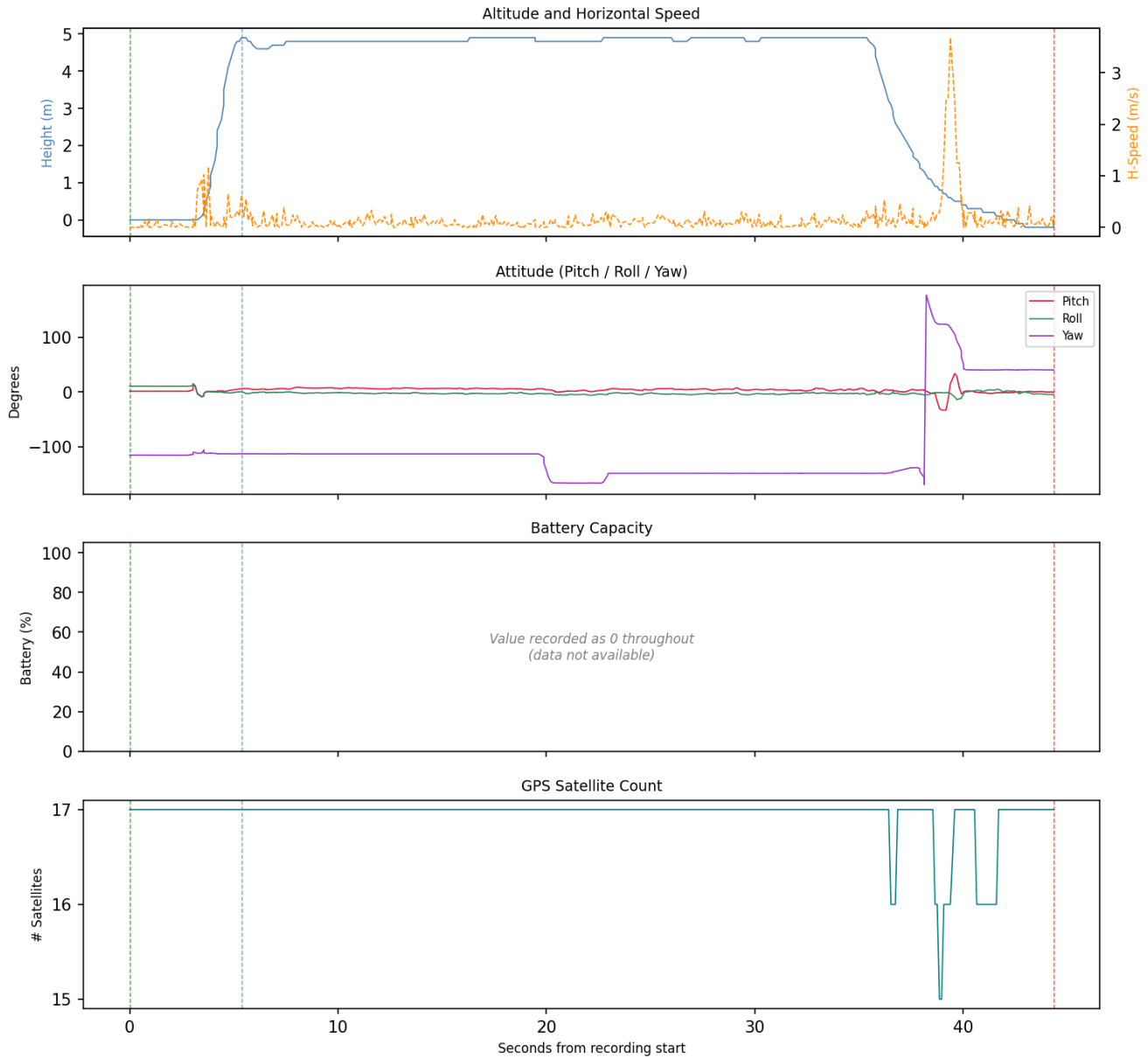
1. EngineStart
2. AssistedTakeoff
3. Sport
4. ForceLanding

Key Events

Timestamp (UTC)	Source	Event	Details
2018-04-19 18:46:27	ctrl_and:fr (v4.2.8)	Log started	SN: 0K1DF2P1BD65JJ; App: 4.2.8; Mavic Air
2018-04-19 18:46:27	ctrl_and:fr (v4.2.8)	Motor turned on	on
2018-04-19 18:46:27	ctrl_and:fr (v4.2.8)	Record mode changed	Starting
2018-04-19 18:46:32	ctrl_and:fr (v4.2.8)	Reached peak height	4.9 m
2018-04-19 18:47:11	ctrl_and:fr (v4.2.8)	Log ended	Dist home: 1.1 m; Height: 0.0 m; Rec: 44.0 s
2018-04-19 18:47:11	ctrl_and:fr (v4.2.8)	Motor turned off	off

Flight Record - Sensor Telemetry

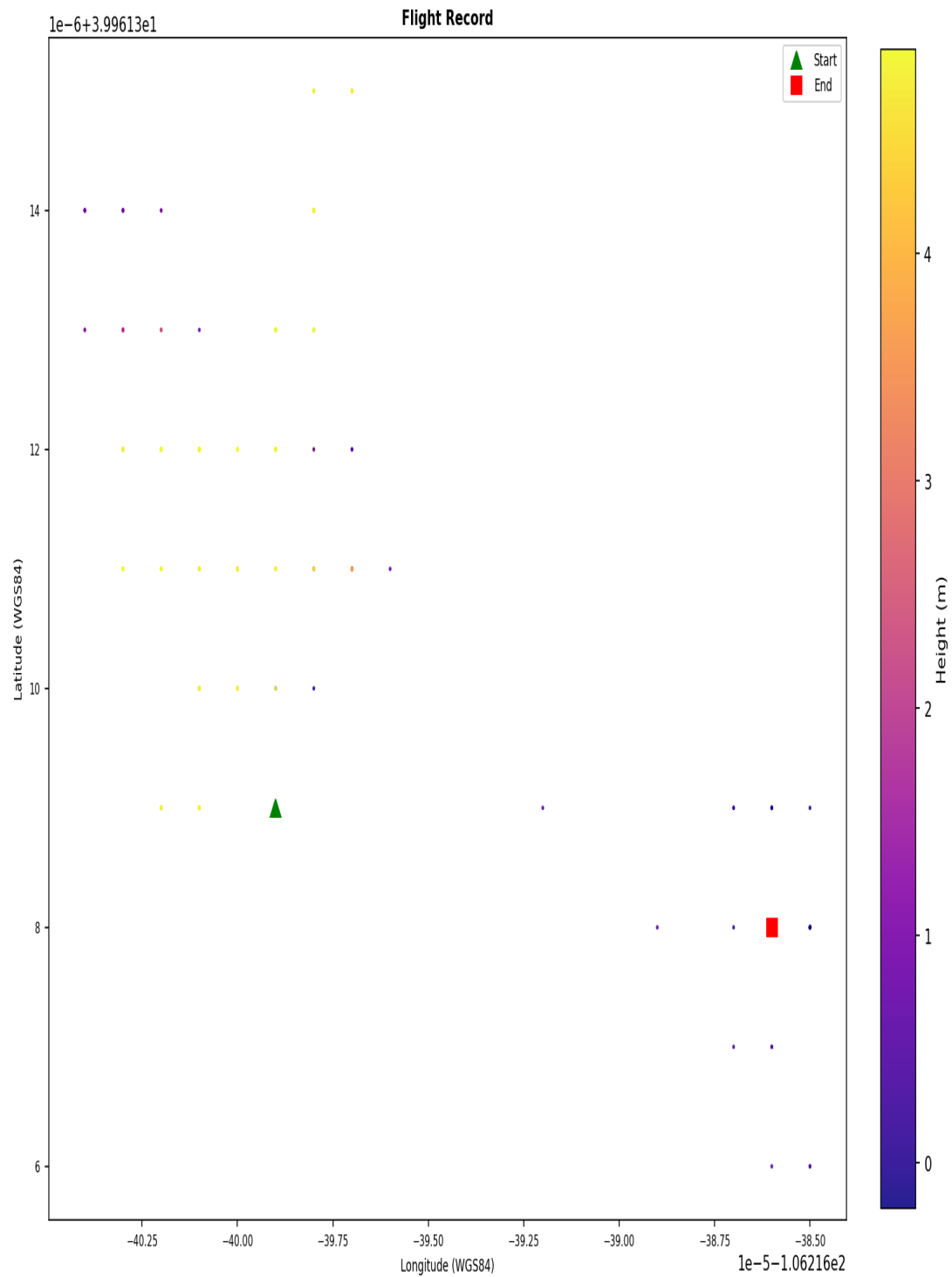
Flight Record Telemetry - flight_01



GPS coordinates are derived from the drone's onboard receiver and represent sensor-recorded positions, not independently verified measurements. A satellite count of 10 or more is associated with good horizontal precision (approximately 1.5-3 m); counts below 4 indicate a poor fix with significantly reduced positional reliability, and those rows are flagged in this report. Low satellite counts may affect the accuracy of the flight path, ground track, and any location-based conclusions drawn from this record.

Flight Ground Track

Flight Ground Track - flight_01



Coordinates are WGS84. No base map. Colour represents relative height (metres).

4. Flight Analysis - Flight 02

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Flight Record SHA-256: 782d29e21aecbbe039d51a68b3085a4ae75626d41c43637c67323bd68000178c
Flight Record Name: DJIFlightRecord_2018-04-19_[13-05-12].csv
Start: 2018-04-19T19:05:12.138000+00:00
End: 2018-04-19T19:05:15.842000+00:00
Duration: 3.7 s

Flight Metrics

Peak Height: -
Distance Travelled: 0.27 m
Video Recording: No
Photo Capture: No
Photos Taken: 0
Recording Duration: 0.0 s
Record Complete: Yes

Fly Modes Observed (Chronological)

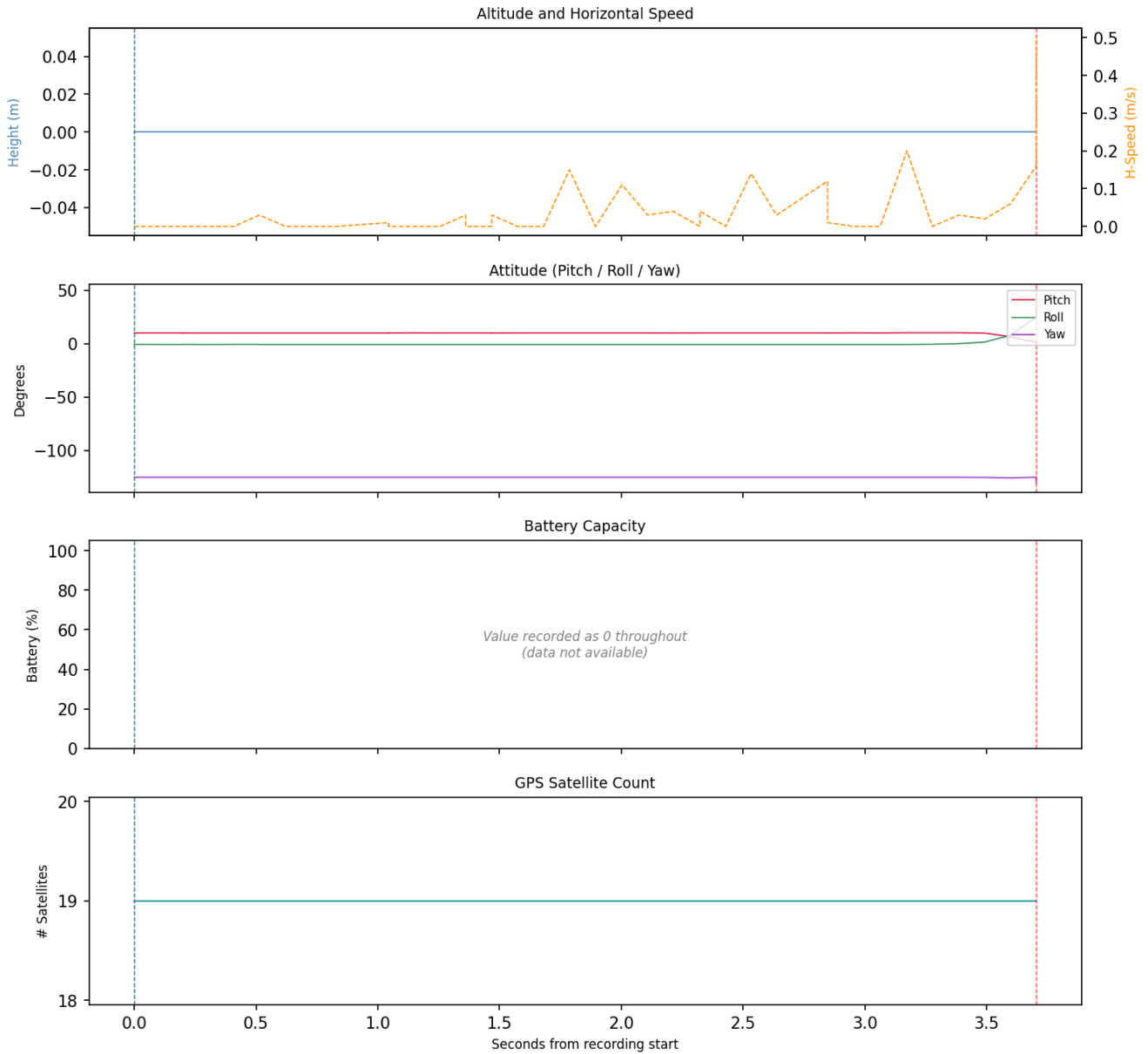
1. EngineStart
2. AssistedTakeoff
3. GPS_Atti

Key Events

Timestamp (UTC)	Source	Event	Details
2018-04-19 19:05:12	ctrl_and:fr (v4.2.8)	Log started	SN: 0K1DF2P1BD; App: 4.2.8; Mavic Air
2018-04-19 19:05:12	ctrl_and:fr (v4.2.8)	Reached peak height	0.0 m
2018-04-19 19:05:12	ctrl_and:fr (v4.2.8)	Motor turned on	on
2018-04-19 19:05:12	ctrl_and:fr (v4.2.8)	Record mode changed	No
2018-04-19 19:05:15	ctrl_and:fr (v4.2.8)	Log ended	Dist home: 0.8 m; Height: 0.0 m; Rec: 0.0 s
2018-04-19 19:05:15	ctrl_and:fr (v4.2.8)	Motor turned off	off

Flight Record - Sensor Telemetry

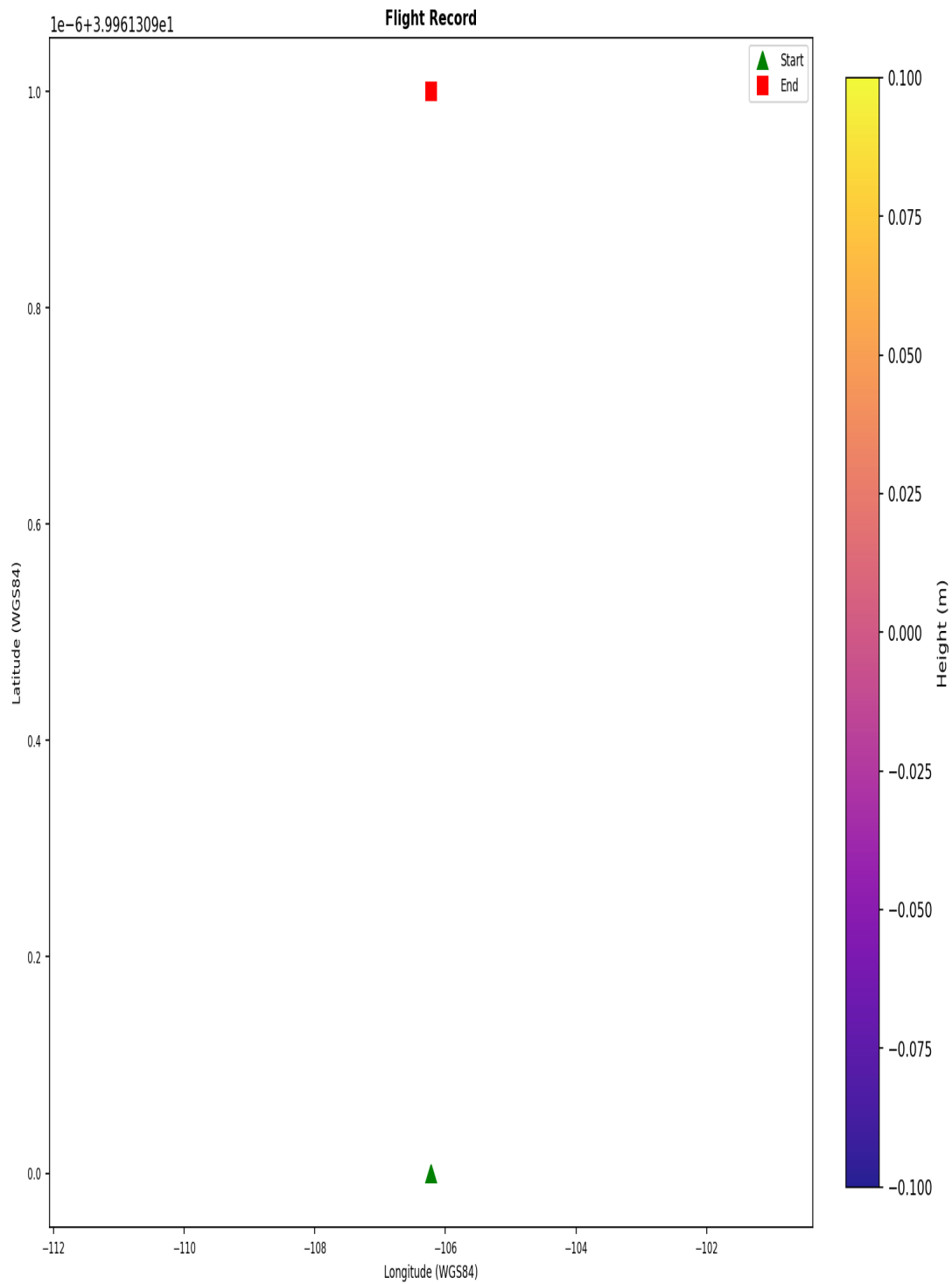
Flight Record Telemetry - flight_02



GPS coordinates are derived from the drone's onboard receiver and represent sensor-recorded positions, not independently verified measurements. A satellite count of 10 or more is associated with good horizontal precision (approximately 1.5-3 m); counts below 4 indicate a poor fix with significantly reduced positional reliability, and those rows are flagged in this report. Low satellite counts may affect the accuracy of the flight path, ground track, and any location-based conclusions drawn from this record.

Flight Ground Track

Flight Ground Track - flight_02



Coordinates are WGS84. No base map. Colour represents relative height (m).

Coordinates are WGS84. No base map. Colour represents relative height (metres).

4. Flight Analysis - Flight 03

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Flight Record SHA-256: a7c66d237b4c95b34433c9179f0642b84ba21ce5a5b4378df0f553821514dbf0
Flight Record Name: DJIFlightRecord_2018-04-19_[13-05-29].csv
Start: 2018-04-19T19:05:29.551000+00:00
End: 2018-04-19T19:13:20.716000+00:00
Duration: 471.2 s

Flight Metrics

Peak Height: 40.8 m
Distance Travelled: 3034.5 m
Video Recording: Yes
Photo Capture: No
Photos Taken: 0
Recording Duration: 264.0 s
Record Complete: No - drone had not returned to home point

Fly Modes Observed (Chronological)

1. GPS_Atti
2. EngineStart
3. AssistedTakeoff
4. Sport
5. ForceLanding
6. AutoLanding

Key Events

Timestamp (UTC)	Source	Event	Details
2018-04-19 19:05:29	ctrl_and:fr (v4.2.8)	Log started	SN: 0K1DF2P1BD; App: 4.2.8; Mavic Air
2018-04-19 19:05:29	ctrl_and:fr (v4.2.8)	Motor turned on	on
2018-04-19 19:08:48	ctrl_and:fr (v4.2.8)	Record mode changed	No
2018-04-19 19:08:56	ctrl_and:fr (v4.2.8)	Record mode changed	Start
2018-04-19 19:08:57	ctrl_and:fr (v4.2.8)	Record mode changed	Starting
2018-04-19 19:09:20	ctrl_and:fr (v4.2.8)	Reached peak height	40.8 m
2018-04-19 19:13:20	ctrl_and:fr (v4.2.8)	Log ended	Dist home: 20.1 m; Height: 0.0 m; Rec: 264.0 s
2018-04-19 19:13:20	ctrl_and:fr (v4.2.8)	Motor turned off	off

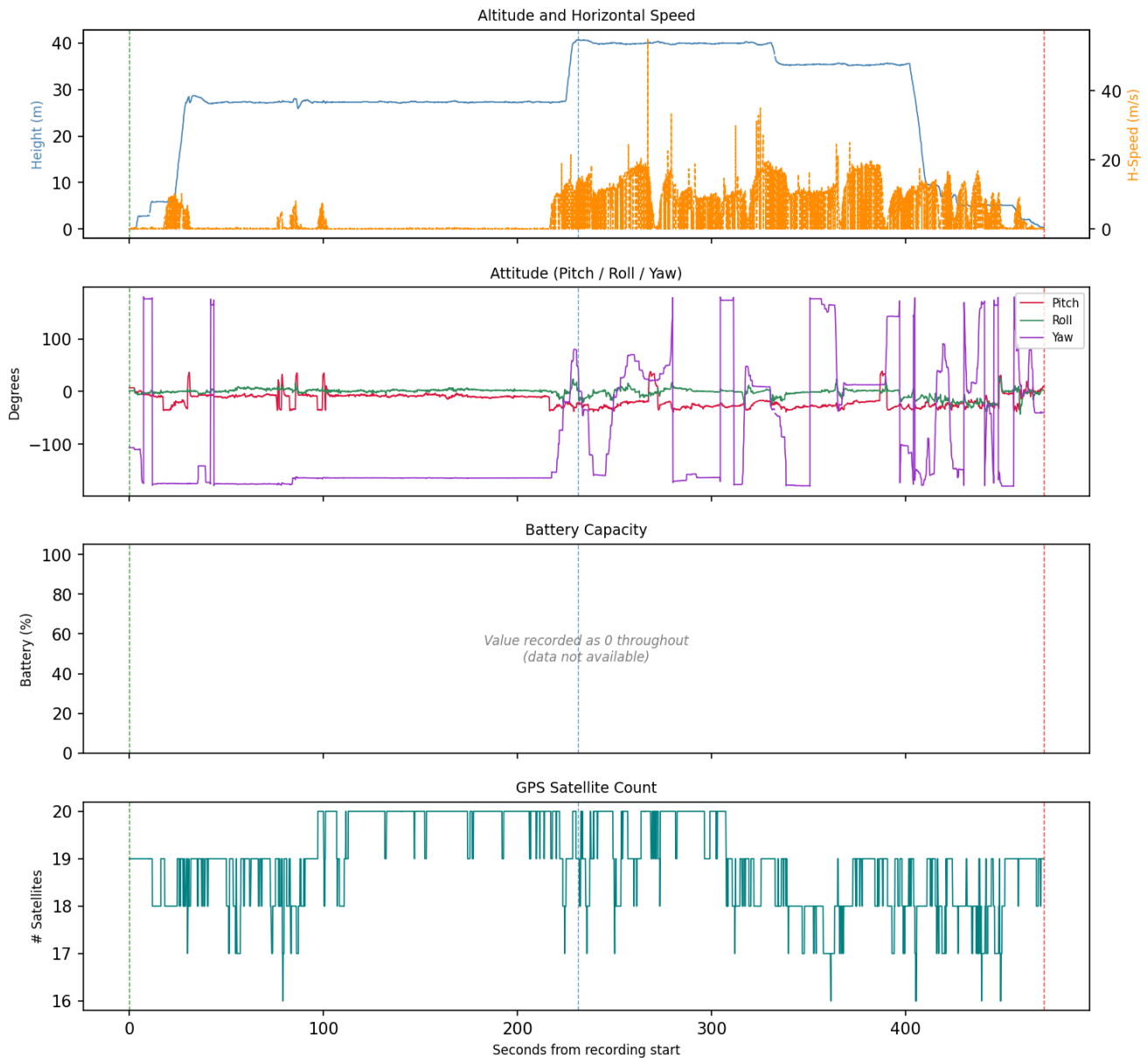
Correlated Artefacts

Possibly correlated (indirect indicators):

controller_android:videos	GPS within flight bounding box (lat 39.9599, lon -106.2167)
p5:1ea0fdbdb3c0f32ceeaf5f9656719c2a15172c0b6d526732fdcf34d4fe33cc4	

Flight Record - Sensor Telemetry

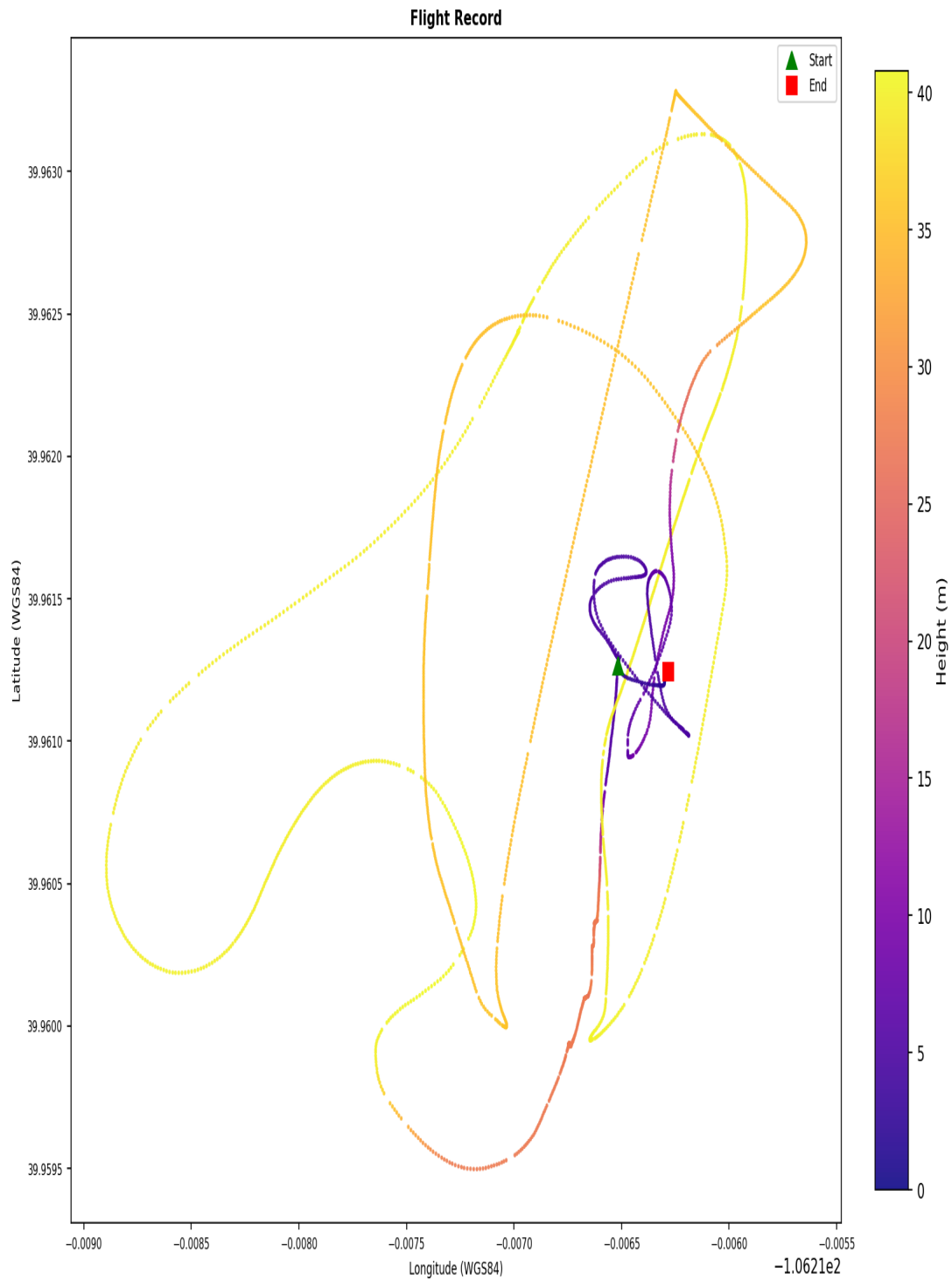
Flight Record Telemetry - flight_03



GPS coordinates are derived from the drone's onboard receiver and represent sensor-recorded positions, not independently verified measurements. A satellite count of 10 or more is associated with good horizontal precision (approximately 1.5-3 m); counts below 4 indicate a poor fix with significantly reduced positional reliability, and those rows are flagged in this report. Low satellite counts may affect the accuracy of the flight path, ground track, and any location-based conclusions drawn from this record.

Flight Ground Track

Flight Ground Track - flight_03



Coordinates are WGS84. No base map. Colour represents relative height (m).

Coordinates are WGS84. No base map. Colour represents relative height (metres).

5. Artefact Coverage

Source Coverage

Source	Detected
controller_android	Yes
controller_ios	No
drone_sd	Yes
drone_flight_storage	Yes

Artefact Counts per Source and Category

Source	Account Data	Databases	Drone Logs	Flight Logs	Flight Records	Images	Videos
controller_android	1	7	0	12	3	3	2
drone_sd	0	0	0	0	0	8	8
drone_flight_storage	0	0	3	0	0	0	0

Data Quality Notes per Category

Drone Logs (3 files):

- 8 file(s) with row anomalies
- 8 file(s) with empty columns
- 8 file(s) with constant-value columns

Flight Logs (12 files):

- 11 file(s) unreadable (binary)
- formats: 1 heading_log

Flight Records (3 files):

- 1 file(s) with row anomalies
- 3 file(s) with empty columns
- 3 file(s) with constant-value columns

Images (11 files):

- 11 file(s) missing (normalisable) EXIF timestamp
- 11 file(s) missing EXIF GPS coordinates

Videos (10 files):

- 1 file(s) contain(s) EXIF GPS coordinates
- 9 file(s) missing (normalisable) EXIF timestamp
- 9 file(s) missing EXIF GPS coordinates

6. Data Quality and Anomalies

Data Quality Observations

- Drone_logs: 8 file(s) with row anomalies detected in telemetry data.
- Drone_logs: 8 file(s) with empty columns - some telemetry channels contain no data.
- Drone_logs: 8 file(s) with constant-value columns - some telemetry channels may be unreliable.
- Flight_logs: 11 file(s) unreadable (binary) - file content could not be parsed.
- Flight_logs: log files present in the following format(s): 1 heading_log.
- Flight_records: 1 file(s) with row anomalies detected in telemetry data.
- Flight_records: 3 file(s) with empty columns - some telemetry channels contain no data.
- Flight_records: 3 file(s) with constant-value columns - some telemetry channels may be unreliable.

Pipeline Anomaly Flags

1. [p2 - controller android - device and backup info]: DeviceInfo.xml not found for android_logical.zip
2. [p2 - controller android - device and backup info]: ApplicationInfo.xml not found for android_logical.zip
3. [p2 - controller ios]: source evidence not found
4. [p3 - controller android - drone logs]: no artefacts found
5. [p3 - controller ios]: source evidence not found
6. [p4 - controller android - account data]: account data parse failed for dji.go.v4.xml
7. [p4 - controller ios]: source evidence not found
8. [p4 - drone flight storage - drone logs]: DatCon reported an error processing: FLY001_1.DAT
9. [p4 - drone flight storage - drone logs]: DatCon reported an error processing: FLY002_1.DAT
10. [p4 - drone flight storage - drone logs]: DatCon reported an error processing: FLY004_1.DAT
11. [p4 - drone flight storage - drone logs]: DatCon reported an error processing: FLY005_1.DAT
12. [p4 - drone flight storage - drone logs]: DatCon reported an error processing: FLY007_1.DAT
13. [p4 - drone flight storage - drone logs]: DatCon reported an error processing: FLY009_1.DAT
14. [p4 - drone flight storage - drone logs]: DatCon reported an error processing: FLY010_1.DAT
15. [p4 - drone flight storage - drone logs]: DatCon reported an error processing: FLY011_1.DAT
16. [p4 - drone flight storage - drone logs]: DatCon reported an error processing: FLY013_1.DAT
17. [p4 - drone flight storage - drone logs]: DatCon reported an error processing: FLY014_1.DAT
18. [p4 - drone flight storage - drone logs]: ExtractDJI only produced GIMBAL processing: DJI_ASSISTANT_EXPORT_FILE_2018-05-07_16-25-09.DAT
19. [p4 - drone flight storage - drone logs]: ExtractDJI only produced GIMBAL processing: DJI_ASSISTANT_EXPORT_FILE_2018-05-07_16-26-14.DAT
20. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY003_1.csv
21. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY006_1.csv
22. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY008_1.csv
23. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY012_1.csv
24. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY015_1.csv
25. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY016_1.csv
26. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY017_1.csv
27. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY018_1.csv

Artefact-Level Anomaly Detail

The following summarises which automated checks fired for each artefact processed in Phase 5. Row counts indicate affected data rows; column lists are truncated to five entries.

Drone Logs

norm_FLY003_1.csv

- missing_gps: 1 row(s) affected
- contains_no_value: 3 column(s) - 4.3.0, Attribute|Value, ConvertDatV3
- contains_constant_value: 2 column(s) - Clock:Tick#, Clock:offsetTime

norm_FLY006_1.csv

- missing_gps: 1 row(s) affected
- contains_no_value: 3 column(s) - 4.3.0, Attribute|Value, ConvertDatV3

- contains_constant_value: 2 column(s) - Clock:Tick#, Clock:offsetTime

norm_FLY008_1.csv

- missing_gps: 1 row(s) affected
- contains_no_value: 3 column(s) - 4.3.0, Attribute|Value, ConvertDatV3
- contains_constant_value: 2 column(s) - Clock:Tick#, Clock:offsetTime

norm_FLY012_1.csv

- missing_gps: 1 row(s) affected
- contains_no_value: 3 column(s) - 4.3.0, Attribute|Value, ConvertDatV3
- contains_constant_value: 2 column(s) - Clock:Tick#, Clock:offsetTime

norm_FLY015_1.csv

- missing_gps: 1 row(s) affected
- contains_no_value: 3 column(s) - 4.3.0, Attribute|Value, ConvertDatV3
- contains_constant_value: 2 column(s) - Clock:Tick#, Clock:offsetTime

norm_FLY016_1.csv

- missing_gps: 1 row(s) affected
- contains_no_value: 3 column(s) - 4.3.0, Attribute|Value, ConvertDatV3
- contains_constant_value: 2 column(s) - Clock:Tick#, Clock:offsetTime

norm_FLY017_1.csv

- missing_gps: 1 row(s) affected
- contains_no_value: 3 column(s) - 4.3.0, Attribute|Value, ConvertDatV3
- contains_constant_value: 2 column(s) - Clock:Tick#, Clock:offsetTime

norm_FLY018_1.csv

- missing_gps: 1 row(s) affected
- contains_no_value: 3 column(s) - 4.3.0, Attribute|Value, ConvertDatV3
- contains_constant_value: 2 column(s) - Clock:Tick#, Clock:offsetTime

Flight Logs

19-04-2018-0K1DF2P1BD65JJ

- format: heading_log | entries: 19

Flight Records

norm_DJIFlightRecord_2018-04-19_[12-46-27].csv

- altitude_negative: 14 row(s) affected
- contains_no_value: 43 column(s) - APP_SER_WARN.warn, APP_WARN.warn, CENTER_BATTERY.connStatus, CENTER_BATTERY.current [A], CENTER_BATTERY.currentCapacity [mAh] (+38 more)
- contains_constant_value: 179 column(s) - APP_GPS.accuracy, APP_GPS.latitude, APP_GPS.longitude, APP_TIP.tip, CAMERA_INFO.cameraType (+174 more)

norm_DJIFlightRecord_2018-04-19_[13-05-12].csv

- contains_no_value: 52 column(s) - APP_SER_WARN.warn, APP_WARN.warn, CENTER_BATTERY.connStatus, CENTER_BATTERY.current [A], CENTER_BATTERY.currentCapacity [mAh] (+47 more)
- contains_constant_value: 188 column(s) - APP_GPS.accuracy, APP_GPS.latitude, APP_GPS.longitude, APP_TIP.tip, CALC.height.running_max [m] (+183 more)

norm_DJIFlightRecord_2018-04-19_[13-05-29].csv

- contains_no_value: 50 column(s) - APP_WARN.warn, CENTER_BATTERY.connStatus, CENTER_BATTERY.current [A], CENTER_BATTERY.currentCapacity [mAh], CENTER_BATTERY.currentPV [V] (+45 more)
- contains_constant_value: 154 column(s) - APP_SER_WARN.warn, CAMERA_INFO.cameraType, CAMERA_INFO.connectState, CAMERA_INFO.enabledPhoto, CAMERA_INFO.encryptStatus (+149 more)

Images

thumb_37464ce56705af10_1524163874000.jpg

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

thumb_8dfde66c5b7928f1_1524164166000.jpg

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

thumb_f02560c4866cf597_1524164234000.jpg

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0001.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0002.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0003.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0004.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0005.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0006.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0007.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0008.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

Videos**2018_04_19_13_08_57.mp4**

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0001.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0002.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0003.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0004.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0005.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0006.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0007.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0008.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

7. Further Investigation Items

The following items were identified by automated analysis as requiring manual investigator follow-up. They represent unresolved questions or limitations that the pipeline cannot resolve automatically.

1. Note: no controller device name found - Device and account information.
2. Drone Serial Number inconsistency detected - conflicting values observed: 0K1DF2P1BD, 0K1DF2P1BD65JJ. Manual verification required.
3. Note: no controller device found - Device and account information.
4. Note: no account e-mail found - Device and account information.
5. Note: no product type found - Device and account information.
6. Note: no firmware version found - Device and account information.
7. Note: no last country code found - Device and account information.
8. Note: no aircraft model code found - Device and account information.
9. Note: no controller serial number found - Device and account information.
10. Note: no last known latitude found - Device and account information.
11. Note: no last known longitude found - Device and account information.
12. Note: no account uid found - Device and account information.
13. Note: no account nickname found - Device and account information.
14. Note: no device uuid found - Device and account information.
15. 3 flight record(s) could not be correlated to a drone log (flight_01, flight_02, flight_03). Independent corroboration is absent for these flight(s).
16. Flight record(s) show signs of incomplete capture (flight_03). The drone may not have returned to the home point before logging stopped. Manual inspection recommended.
17. 7 database file(s) contain records and should be examined for relevant artefacts.

Appendix A - Extracted Artefact Hash Manifest

Each artefact extracted in Phase 3 is listed below with its full SHA-256 and SHA-1 cryptographic hashes. These values can be used to independently verify the integrity of every extracted file.

dji.go.v4.xml	Account Data	controller_android	extract_logical
SHA-256:	62cc6a7fc9d83bcfd63c7cba006d12880df3d9cec2a261dc9ccb47f317735678		
SHA-1:	d28ede46ad8fdf1e2cb9b78d9e80a171f47be6bb		

dafile.db	Databases	controller_android	extract_logical
SHA-256:	80a63197c66ee5a35b099c89318be41d55b04188fdb20d74a3bdea1bc7ab5fd7		
SHA-1:	92f614196df7215a7bbb5ca1c411f948ec21d6a6		

dji.db	Databases	controller_android	extract_logical
SHA-256:	5fd60661d3ea4ec8d158a4fc9a5bce210a40bb417ff78d20dd4dc189a9b5e497		
SHA-1:	b86a0eee9f853c749f3552c83ac452ec516d489d		

flysafe_areas_djigo.db	Databases	controller_android	extract_logical
SHA-256:	a21513f2ae3bebe5c80a26c654403e108bef51629a2201a47b8838434abfa70a		
SHA-1:	674c8f1245f5c9c15dc02fffa1ea2690d688f591		

flysafe_areas_djigo_1.db	Databases	controller_android	extract_logical
SHA-256:	d9164be112208833fe06583920459508675742a54e3ca70d358789ab362c7d3c		
SHA-1:	936e7b4a02e38dc7cd2b2d4cb7be8280096b17bd		

flysafe_polygon_1860.db	Databases	controller_android	extract_logical
SHA-256:	bad78ebc14dcc7b63d2b807d0ced2d25d129d8bbc3d93285a6d986f06e0e8cd4		
SHA-1:	a2685bcff9b1ff7e82b4400f8e59b9ca3ff82ebb		

logdb.db	Databases	controller_android	extract_logical
SHA-256:	05d81638a36413b64454138740e09fd88d2112b8fec7f8b28048c43ba526fe8f		
SHA-1:	8813ec28e3133d5317fe699c9f0e9d8c6ac9183e		

mbgl-offline.db	Databases	controller_android	extract_logical
SHA-256:	6bf96d99a348d74dff61dc708622c5a398a238ca3ae26e34a0cddf94b037846e		
SHA-1:	a50e6f1d3ee3b358ba409b6ad78d292227ebecda		

DJI_ASSISTANT_EXPORT_FILE_2018-05-07_	Drone Logs	drone_flight_storage	extract_logical
SHA-256:	3aac7c24fae69dfd6d01dab7fed52f11cfec9ef2b5249becd25722619cfb3adf		
SHA-1:	36d5d41c2f3445a5520650f4d18dc1875bbc3c60		

DJI_ASSISTANT_EXPORT_FILE_2018-05-07_	Drone Logs	drone_flight_storage	extract_logical
SHA-256:	ffed36fd36064df9c243d039f152942b008ca5a8e4088144a110900ec052d752		
SHA-1:	fa38e84fc473affc6429d851d276a45f86c5d38a		

DJI_ASSISTANT_EXPORT_FILE_2018-05-07_	Drone Logs	drone_flight_storage	extract_logical
SHA-256:	6d59d245215382ac6326cd78b47955141402ea95866813a980171dcbda5300b4		
SHA-1:	467aa0a825395c59c64f0204a3bf8f2b697d2dd2		

1524163431241	Flight Logs	controller_android	extract_logical
SHA-256:	281d8cd75b0210365c0c506fa3aab08d43e31f92783ff2e11a6312a55f9f881a		
SHA-1:	7b48184096e018a02dd49f33275aa75c6397e06a		

1524163523963		Flight Logs	controller_android	extract_logical
SHA-256:	5c4f3cd0dabc2bfe4ade97e151846e1953ef7bcbf5efed95fe7156cbea056bd1			
SHA-1:	88adb52cc45d24589566390e9b35994cb38e9644			

1524163893736		Flight Logs	controller_android	extract_logical
SHA-256:	a1791032c55e3f01cab002d749f1b68c6b6eb25962f190b9fdcc094e1fec1918			
SHA-1:	8170f27f31033e68663e1fc21468326618b4c2f7			

1524164136352		Flight Logs	controller_android	extract_logical
SHA-256:	e92cb52942c0c9bf36ed8a0b0414103ca95edb2c536e314af0bd96630d5673cc			
SHA-1:	70f178329ee8edf0d36eb6d075c7c503607678bc			

1524164315248		Flight Logs	controller_android	extract_logical
SHA-256:	bb8afa7b08d841d70e66c07da24e29cb8e8668597d45a8ff1c94a96db3db5919			
SHA-1:	8bcc364a3a8ecb39729627cf42c3ac53b28fffc			

1524164350973		Flight Logs	controller_android	extract_logical
SHA-256:	3e429509601496e8accd3454afdeb6937820299a7bb669939a14c538c1491420			
SHA-1:	ef8eeddea9cbd80160ed25ef4a650e79acba69e			

1524165212553		Flight Logs	controller_android	extract_logical
SHA-256:	7d54165053b31e932de05bc1659195cf0653e58f9869f14c695c48a78246f015			
SHA-1:	87249500f5cae1ea87eb8223b0a0cf2405771598			

19-04-2018-0K1DF2P1BD65JJ		Flight Logs	controller_android	extract_logical
SHA-256:	5bd2f43c4cdfc4c4402cc8e3f5195b418ec01858d27d0ca7ea3ada7e42e62dba			
SHA-1:	70f29ff5d2b0512eb8b1a5470e9c478eeaf436a9			

MANIFEST-000002		Flight Logs	controller_android	extract_logical
SHA-256:	8730d99ac897d5e850264f48be2953f9facd88181df474b13e2b661c200fdbef			
SHA-1:	64e526a95663d39481654a2bbb5261901472ca4b			

MANIFEST-000002_1		Flight Logs	controller_android	extract_logical
SHA-256:	32ee4bfdb8e198e776faaa60863d439e3617291c48dfb41833b2a5d761936542			
SHA-1:	2fa0742ce2d68599b419a387bfaf97aa3f27948e			

MANIFEST-000004		Flight Logs	controller_android	extract_logical
SHA-256:	2161b83a03242941c198406a4324bfaf579fdd568ecfeb55ce31e9ae68046139			
SHA-1:	5a8f234dd240f6f7bb5409c8de192ed76d3936ac			

MANIFEST-000004_1		Flight Logs	controller_android	extract_logical
SHA-256:	44eb4e2dd2e1c0fdcb92467ed407c746e43ff97b12db3b79f6aa505e080a7fc4			
SHA-1:	21ca4cbc772d8904be9be3e2247a4b479c15cb41			

DJIFlightRecord_2018-04-19_[12-46-27].txt		Flight Records	controller_android	extract_logical
SHA-256:	8365a1f3dea62bd2a359819de198387b04f77dd9d5b760514197e48dc00a6607			
SHA-1:	d336e7a29e2b797b1b1aeadd113e0160d83404eb5			

DJIFlightRecord_2018-04-19_[13-05-12].txt		Flight Records	controller_android	extract_logical
SHA-256:	18a29733318941302357eaf543e3c222480ef254f84bbb19163f9d112b89adb6			
SHA-1:	7386e0097929a29d7325aa7b23e464e00b2cc4d8			

DJIFlightRecord_2018-04-19_[13-05-29].txt		Flight Records	controller_android	extract_logical
SHA-256:	db71b31cbd6aa7f7a693d86a7aea2765e9e648aa6bbd59554db87e9fc6f9cb4a			
SHA-1:	c221d7a9cd0c3f5655a3d687e2a2e9309d375291			

thumb_37464ce56705af10_1524163874000.jpg		Images	controller_android	extract_logical
SHA-256:	59606cacde0c4c6f84f055ff0cdc19dcc4e6fe57e2749656374d0fe3fa003bb1			
SHA-1:	72b6cbfc7f14843610c2a5ba122bc98a5e72ffc7			

thumb_8dfde66c5b7928f1_1524164166000.jpg		Images	controller_android	extract_logical
SHA-256:	30762f01607f3bbf92afb086e6c6b752b43e3ee55e4a1b446e6ca8b934272d07			
SHA-1:	7b56b31db5ef34bd0fe7cebb533c6817d7a12e49			

thumb_f02560c4866cf597_1524164234000.jpg		Images	controller_android	extract_logical
SHA-256:	55aecb7dac3be801aaed3cd31efe967d6adf603a6f99838685fe5224ceb17b93			
SHA-1:	ce63c216ec1bc3dced1a0c840511eaf37b382725			

DJI_0001.THM		Images	drone_sd	extract_physical
SHA-256:	c68e4a22ce22a014c7b50525589506af23c439c5ce169a1a2df2c04e6dfa42cc			
SHA-1:	b14dbe44ef37fdb6cdf53a2664b7ec5308850865			

DJI_0002.THM		Images	drone_sd	extract_physical
SHA-256:	4f59b9739fe08082c28edfc558514096cf836bf4ac8b4f225bddcf056abd76d6			
SHA-1:	d31f66c08ed26978adfacc04184430fd0959730fc			

DJI_0003.THM		Images	drone_sd	extract_physical
SHA-256:	4554c5231b411035308eff355a71d338f39c67896d4f145b11fd337ada50b63e			
SHA-1:	a4d039b7c480574b73b1c4ae427d2a8f2f501873			

DJI_0004.THM		Images	drone_sd	extract_physical
SHA-256:	6fe85cfac77e1b6fab2a3c0d8830785b0dc45a633cd644c0558c76a5634220d1			
SHA-1:	21a3202204e6bb6c0fb1319c8be760bfd56ddd0e			

DJI_0005.THM		Images	drone_sd	extract_physical
SHA-256:	bb1500881fd04845d8b249455f84be01f2e2f840759d50bedba39d234635d3ea			
SHA-1:	f1f6dbbf74732745131a6b777ef22d7eda2a24f5			

DJI_0006.THM		Images	drone_sd	extract_physical
SHA-256:	bc16e0bf4718d299de54775479840a91bfe6527974d77057a9d5f48db4db0f5d			
SHA-1:	b0b50a823ef2b86456d065685f0bf3be75f05016			

DJI_0007.THM		Images	drone_sd	extract_physical
SHA-256:	b74120c05dc189c51d5b98b6a9b45f8df6f179ce84cca3ab70e362ea9cc0aca0			
SHA-1:	3988c7703061dad1426d2974bdbcbdb6a0e1cdb2			

DJI_0008.THM		Images	drone_sd	extract_physical
SHA-256:	3abcb6c3d8d9aa8882adda4f3d511bcea72a6cc1850c71450e4f1a71d706278			
SHA-1:	4ae82e4150e448307ddab7f522a0e50c6a6887eb			

2018_04_19_13_08_57.info		Videos	controller_android	extract_logical
SHA-256:	1ea0fdbdb3c0f32ceeaf5f9656719c2a15172c0b6d526732fdcf34d4fe33cc4			
SHA-1:	830faa51950ee2fdc1c55c1d99dbd64e5d28ba6c			

2018_04_19_13_08_57.mp4		Videos	controller_android	extract_logical
SHA-256:	a72d027992b02648297751f369b2130a61e1e998827681f84a8dae3d525d36d9			
SHA-1:	76278f3d257b0d5e4a2289365758783591548d15			

DJI_0001.MP4		Videos	drone_sd	extract_physical
SHA-256:	598bd1a0b57233e20c519b98c09da7c90f52b544811a35da57e471055c8fbca7			
SHA-1:	a728af1b38100789ab6bdb5a57ff4d981e45f3e2			

DJI_0002.MP4		Videos	drone_sd	extract_physical
SHA-256:	383f9571802bbd28132baee00aab439321eb20856ba0c71df02ed38f70dc6940			
SHA-1:	53cb5109651a3e58b06c42116e329730d4ae345c			

DJI_0003.MP4		Videos	drone_sd	extract_physical
SHA-256:	081fed0976f0b8a58aa73f2b407acbcf7674f717dc33f55d8d10cdad010bef24			
SHA-1:	7dde5b92e7746bc9ee39fcd86c86f80395248030			

DJI_0004.MP4		Videos	drone_sd	extract_physical
SHA-256:	3488c4a86596c628ca67ef7714735118410c5c47b18c01aa49e19aef630bea21			
SHA-1:	8cd037d2b4a7df7493026443b5573bab407df8fc			

DJI_0005.MP4		Videos	drone_sd	extract_physical
SHA-256:	8170c7543d17db18bb64b83829f5365d87e39643616d106cc1f48838707cefd8			
SHA-1:	068d9f8d0085915bfea01b6e099c21af3e0b72d1			

DJI_0006.MP4		Videos	drone_sd	extract_physical
SHA-256:	412a809e70a5fed4f284d100137ed7bf625c8cf0e55741bb97c088f7b5dd6b61			
SHA-1:	e80a2d1519cb92438eb6a8f868906394bffed86c			

DJI_0007.MP4		Videos	drone_sd	extract_physical
SHA-256:	f7e312d10fcd1c35b5324b481f2e37db3af339272bfb53519bbd3de967f6060c			
SHA-1:	8d20e58e134343b9a6cd4b8bce218b2086aa8f6f			

DJI_0008.MP4		Videos	drone_sd	extract_physical
SHA-256:	dc93b87bf71e4ca0b3354db11f1cdf362c2c89e213f69479c2013ff9666ff9cc			
SHA-1:	7e89b45e4057ca21e9c8bb304b146ecfca6d57c1			

Appendix B - Evidence Derivation Summary

This table summarises how many artefacts were derived at each pipeline phase, per evidence source and category. It demonstrates the processing chain from extracted artefacts (P3) through decoded outputs (P4) to normalised analysis-ready files (P5).

Controller Android

Category	P3 Extracted	P4 Decoded	P5 Normalised
Account Data	1	0	0
Databases	7	0	0
Flight Logs	12	0	0
Flight Records	3	3	3
Images	3	3	0
Videos	2	2	0

Drone Sd

Category	P3 Extracted	P4 Decoded	P5 Normalised
Images	8	8	8
Videos	8	8	0

Drone Flight Storage

Category	P3 Extracted	P4 Decoded	P5 Normalised
Drone Logs	3	42	8

Appendix C - Tool Invocation Log

All external tool invocations executed during this analysis are recorded below, in chronological order. Each entry includes the timestamp, tool name, version, return code, full command, and output files produced.

2026-06-18 12:37:30 mmls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: mmls.exe df047_external.E01

2026-06-18 12:37:43 fls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: fls.exe -r -p -o 8192 df047_external.E01

2026-06-18 12:39:22 parser_android v- rc=n/a

cmd: android_logical.zip C:\Users\Floris\Documents\dfdof_output\ft-df050-2018-android\p2_image_parsing\controller_android_parsed
out: controller_android_parsed

2026-06-18 12:39:38 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df047_external.E01 1357829
out: DJI_0001.MP4

2026-06-18 12:40:01 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df047_external.E01 1357830
out: DJI_0002.MP4

2026-06-18 12:40:25 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df047_external.E01 1357831
out: DJI_0003.MP4

2026-06-18 12:40:38 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df047_external.E01 1357832
out: DJI_0004.MP4

2026-06-18 12:40:53 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df047_external.E01 1357833
out: DJI_0005.MP4

2026-06-18 12:41:16 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df047_external.E01 1357834
out: DJI_0006.MP4

2026-06-18 12:41:39 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df047_external.E01 1357835
out: DJI_0007.MP4

2026-06-18 12:41:55 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df047_external.E01 1357836
out: DJI_0008.MP4

2026-06-18 12:42:00 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df047_external.E01 1358853
out: DJI_0001.THM

2026-06-18 12:42:00 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df047_external.E01 1358854
out: DJI_0002.THM

2026-06-18 12:42:01 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df047_external.E01 1358855
out: DJI_0003.THM

2026-06-18 12:42:01 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df047_external.E01 1358856
out: DJI_0004.THM

2026-06-18 12:42:01 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df047_external.E01 1358857
out: DJI_0005.THM

2026-06-18 12:42:01 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df047_external.E01 1358858
out: DJI_0006.THM

2026-06-18 12:42:01 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df047_external.E01 1358859
out: DJI_0007.THM

2026-06-18 12:42:01 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df047_external.E01 1358860
out: DJI_0008.THM

2026-06-18 12:42:21 txtlogtocsvtool v2018-06-11 rc=0

cmd: TXTlogToCSVtool.exe DJIFlightRecord_2018-04-19_[12-46-27].txt DJIFlightRecord_2018-04-19_[12-46-27].csv
out: DJIFlightRecord_2018-04-19_[12-46-27].csv

2026-06-18 12:42:22 txtlogtocsvtool v2018-06-11 rc=0

cmd: TXTlogToCSVtool.exe DJIFlightRecord_2018-04-19_[13-05-12].txt DJIFlightRecord_2018-04-19_[13-05-12].csv
out: DJIFlightRecord_2018-04-19_[13-05-12].csv

2026-06-18 12:42:22 txtlogtocsvtool v2018-06-11 rc=0

cmd: TXTlogToCSVtool.exe DJIFlightRecord_2018-04-19_[13-05-29].txt DJIFlightRecord_2018-04-19_[13-05-29].csv
out: DJIFlightRecord_2018-04-19_[13-05-29].csv

2026-06-18 12:42:23 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n thumb_37464ce56705af10_1524163874000.jpg
out: thumb_37464ce56705af10_1524163874000_exif.json

2026-06-18 12:42:23 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n thumb_8dfde66c5b7928f1_1524164166000.jpg
out: thumb_8dfde66c5b7928f1_1524164166000_exif.json

2026-06-18 12:42:23 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n thumb_f02560c4866cf597_1524164234000.jpg
out: thumb_f02560c4866cf597_1524164234000_exif.json

2026-06-18 12:42:24 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2018_04_19_13_08_57.mp4
out: 2018_04_19_13_08_57_exif.json

2026-06-18 12:42:24 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.THM
out: DJI_0001_exif.json

2026-06-18 12:42:24 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0002.THM
out: DJI_0002_exif.json

2026-06-18 12:42:25 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.THM
out: DJI_0003_exif.json

2026-06-18 12:42:25 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0004.THM
out: DJI_0004_exif.json

2026-06-18 12:42:25 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0005.THM
out: DJI_0005_exif.json

2026-06-18 12:42:25 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0006.THM

out: DJI_0006_exif.json

2026-06-18 12:42:26 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0007.THM

out: DJI_0007_exif.json

2026-06-18 12:42:26 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0008.THM

out: DJI_0008_exif.json

2026-06-18 12:42:26 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.MP4

out: DJI_0001_exif.json

2026-06-18 12:42:27 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0002.MP4

out: DJI_0002_exif.json

2026-06-18 12:42:27 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.MP4

out: DJI_0003_exif.json

2026-06-18 12:42:28 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0004.MP4

out: DJI_0004_exif.json

2026-06-18 12:42:28 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0005.MP4

out: DJI_0005_exif.json

2026-06-18 12:42:29 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0006.MP4

out: DJI_0006_exif.json

2026-06-18 12:42:29 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0007.MP4

out: DJI_0007_exif.json

2026-06-18 12:42:29 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0008.MP4

out: DJI_0008_exif.json

2026-06-18 12:43:12 extractdji v1.4.3 rc=n/a

cmd: ExtractDJI.exe DJI_ASSISTANT_EXPORT_FILE_2018-05-07_16-22-54.DAT

C:\Users\Floris\Documents\dfdof_output\ft-df050-2018-android\p4_decision_and_orchestration\drone_flight_storage\drone_logs

out: FLY001.DAT, FLY002.DAT, FLY003.DAT, FLY004.DAT (+14 more)

2026-06-18 12:44:14 datcon v4.3.0 rc=1

cmd: DatCon.4.3.0.exe FLY001_1.DAT

2026-06-18 12:44:35 datcon v4.3.0 rc=1

cmd: DatCon.4.3.0.exe FLY002_1.DAT

2026-06-18 12:44:47 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY003_1.DAT

out: FLY003_1.csv, FLY003_1.DAT, FLY003_1.kml

2026-06-18 12:44:57 datcon v4.3.0 rc=1

cmd: DatCon.4.3.0.exe FLY004_1.DAT

2026-06-18 12:45:05 datcon v4.3.0 rc=1

cmd: DatCon.4.3.0.exe FLY005_1.DAT

2026-06-18 12:45:16 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY006_1.DAT

out: FLY006_1.csv, FLY006_1.DAT, FLY006_1.kml

2026-06-18 12:45:24 datcon v4.3.0 rc=1

cmd: DatCon.4.3.0.exe FLY007_1.DAT

2026-06-18 12:45:34 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY008_1.DAT

out: FLY008_1.csv, FLY008_1.DAT, FLY008_1.kml

2026-06-18 12:45:41 datcon v4.3.0 rc=1

cmd: DatCon.4.3.0.exe FLY009_1.DAT

2026-06-18 12:45:47 datcon v4.3.0 rc=1

cmd: DatCon.4.3.0.exe FLY010_1.DAT

2026-06-18 12:45:55 datcon v4.3.0 rc=1

cmd: DatCon.4.3.0.exe FLY011_1.DAT

2026-06-18 12:46:05 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY012_1.DAT

out: FLY012_1.csv, FLY012_1.DAT, FLY012_1.kml

2026-06-18 12:46:12 datcon v4.3.0 rc=1

cmd: DatCon.4.3.0.exe FLY013_1.DAT

2026-06-18 12:46:19 datcon v4.3.0 rc=1

cmd: DatCon.4.3.0.exe FLY014_1.DAT

2026-06-18 12:46:29 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY015_1.DAT

out: FLY015_1.csv, FLY015_1.DAT, FLY015_1.kml

2026-06-18 12:46:40 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY016_1.DAT

out: FLY016_1.csv, FLY016_1.DAT, FLY016_1.kml

2026-06-18 12:46:49 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY017_1.DAT

out: FLY017_1.csv, FLY017_1.DAT, FLY017_1.kml

2026-06-18 12:47:09 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY018_1.DAT

out: FLY018_1.csv, FLY018_1.DAT, FLY018_1.kml

2026-06-18 12:47:44 extractdji v1.4.3 rc=1

cmd: ExtractDJI.exe DJI_ASSISTANT_EXPORT_FILE_2018-05-07_16-25-09.DAT
C:\Users\Floris\Documents\dfdif_output\ft-df050-2018-android\p4_decision_and_orchestration\drone_flight_storage\drone_logs

2026-06-18 12:48:09 extractdji v1.4.3 rc=1

cmd: ExtractDJI.exe DJI_ASSISTANT_EXPORT_FILE_2018-05-07_16-26-14.DAT
C:\Users\Floris\Documents\dfdif_output\ft-df050-2018-android\p4_decision_and_orchestration\drone_flight_storage\drone_logs

Appendix D - DFDOF Proof of Concept Overview

The Drone Forensic Decision and Orchestration Framework (DFDOF) is an automated 8-phase forensic pipeline for DJI drone evidence. It ingests images of the controller, drone flight storage and SD cards, then processes them into a structured analysis and report.

P1 Provenance and Integrity

Classifies input files (ZIP/E01/.001) by content signatures. Computes SHA-256 and SHA-1 for all input evidence. Presents a source classification summary for operator confirmation before processing begins.

Input: evidence directory. Output: classified evidence objects with hashes.

P2 Image Parsing

Makes all sources search-ready. Decrypts iOS backups into a domain directory tree. Extracts Android device metadata (DeviceInfo.xml, ApplicationInfo.xml, packages.list). Produces backup_info.json with device identifiers and installed DJI applications.

Input: raw evidence files. Output: parsed directory structures, backup_info.json.

P3 Artefact Extraction

Root-driven category extraction. Discovers DJI app roots by bundle-ID segment matching. Recurses into configured per-platform category paths and filters by extension. Rejects empty files. Each artefact is wrapped in an Evidence object with a parent hash linking it to its source.

Input: parsed directories. Output: categorised artefacts (flight records, drone logs, flight logs, databases, images, videos, account data).

P4 Decision and Orchestration

Dispatches each artefact category to the appropriate specialist tool: DatCon for .DAT drone logs (produces CSV + KML), TXTlogToCSV for flight record .txt files, ExifTool for images and videos, Python parsers for account data (.plist/.xml). Computes hashes for all tool outputs.

Input: extracted artefacts. Output: decoded CSVs, KML files, EXIF JSON, account data observations.

P5 Normalisation and Anomaly Checking

Normalises all CSVs to UTC timestamps and a unified schema. Applies 9 shared telemetry anomaly checks (timestamp regression/gaps, zero coordinates, GPS quality, altitude spikes, motor-airborne-off) plus format-specific checks for flight records (battery voltage, temperature) and drone logs (quaternion, hDOP). Detects empty databases, unreadable flight logs, and missing (normalisable) EXIF timestamps.

Input: decoded artefacts. Output: normalised CSVs, anomaly observations.

P6 Multi-Source Correlation

Correlates flight records with drone logs using two criteria: (1) temporal overlap ≥ 60 s AND $\geq 90\%$ of the shorter flight's duration; (2) median haversine distance of nearest-neighbour GPS pairs ≤ 25 m with ≥ 5 matched pairs. Deduplicates FR candidates when multiple apps recorded the same flight. Builds a unified per-flight event timeline from all matched sources. Also correlates media files by EXIF timestamp, GPS bounding box, and video duration. In the correlation summary: Overlap is the temporal overlap as a percentage of the shorter flight's duration; GPS Match Rate is the fraction of flight-record GPS rows that found a spatially matched drone-log row within a ± 2 s window.

Input: normalised CSVs. Output: timeline_flightXX.json per flight.

P7 Analysis and Validation

Aggregates results from all prior phases into a structured analysis. Resolves device identity (drone serial, model, app version, account email) across sources with confidence levels (controller-only / corroborated / inconsistent). Computes per-flight metrics (peak height, media capture, record completeness). Produces coverage scores and a grouped conclusions structure with a dedicated further_investigation section.

Input: all phase outputs. Output: analysis dict with conclusions in state.json.

P8 Automated Reporting

Generates this PDF forensic baseline report from all prior phase outputs. Produces sensor telemetry plots (altitude, speed, attitude, battery, GPS) from normalised CSVs, a dual-source flight track plot, and the structured appendices. All figures are also saved as standalone PNG files.

Input: state.json (all phase outputs). Output: PDF report + PNG plot files.

Disclaimer

DFDOF was developed by Floris Krijger, MSc Security and Network Engineering, University of Amsterdam, as a Master's thesis project (1 April 2026 - 30 June 2026). This proof of concept has been developed and tested on a targeted subset of the NIST VTO forensic drone dataset and is not a finished, production-ready system. Before use in any operational or legal context, further testing, validation across additional drone models and evidence types, and robustness improvements are required. The authors accept no liability for conclusions drawn from automated pipeline output without independent expert verification.